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Assignment 7

Assignment Overview:

The goal of this assignment is to research Solaris and Trusted Extensions. This task involves creating a security design related to Trusted Extensions for a new government intelligence agency that must ensure there is no read up and no write down between the classifications. You will design the labels and roles for the system that must support multilevel security. The system will only have one user accessing the system who is a security analyst that will analyze intelligence reports. The final deliverable will be a report of your findings including descriptions from the labels you created and role. Further details are included below.

Main Tasks:

This project will encompass the following tasks:

- 1. Research Solaris Trusted Extensions and explain the key components of Zones and Roles.**

Solaris Trusted Extensions uses zones and roles to help secure the system and control access. Zones are separate environments inside one system that are labeled with different classification levels, such as Confidential or Secret. According to Oracle, labeled zones are used to isolate data and processes based on their security level, which helps prevent information from mixing between classifications. Each zone only contains data that matches its label, and users can only access zones based on their clearance level. There is also a global zone, which is mainly used for system administration and has higher privileges than the other zones.

Roles are used to manage administrative access in a more secure way. Instead of giving a user full administrator privileges all the time, Trusted Extensions allows users to assume roles only when needed. Oracle explains that roles are special accounts used for performing administrative tasks, which helps separate normal user activities from system management. For example, a user can log in normally to do their work, but switch to an administrative role when they need to make changes to the system. This helps reduce security risks because users are not always operating with elevated privileges.

- 2. Assign the user a clearance level. (Example Classifications: Unclassified, Controlled Unclassified, Confidential, Secret, and Top Secret).**

For this system, I would assign the security analyst a Secret clearance level. I chose Secret because the user works as a security analyst for a government intelligence agency and needs access to sensitive intelligence reports. With a Secret clearance, the analyst can access information labeled Unclassified, Controlled Unclassified Information, Confidential, and Secret. However, the analyst should not be allowed to access Top Secret information because that is above their clearance level.

- 3. Design a theoretical labeled zone architecture using at least two classification levels from the list above for the new government agency. Describe what the system should do when an access attempt or write operation occurs from the user between the classification level selected.**

Zone	Classification Level	Purpose
cui_zone	Controlled Unclassified Information	For sensitive but lower-level agency information
confidential_zone	Confidential	For internal intelligence reports
secret_zone	Secret	For sensitive intelligence reports

The security analyst would mainly work in the **secret_zone** because their clearance level is Secret. This allows them to access both Secret information and lower-level data when needed.

Access Control: The system should follow the rules of no read up and no write down.

- If the analyst is working in the **confidential_zone** and tries to open a file from the **secret_zone**, the system should deny access. This is because Secret is higher than Confidential, and the user should not be able to “read up.”
- If the analyst is working in the **secret_zone** and tries to copy or save a Secret report into the **confidential_zone** or **cui_zone**, the system should block the action. This prevents Secret information from being written down to a lower classification level.

In simple terms, the analyst can read information at their level or lower, but they cannot read higher-level information or move higher-level information into a lower-level zone.

- 4. Design at least one administrative role and describe the conditions for the user to be allowed to assume the role.**

For this system, I would create a security administrator role. This role would be responsible for managing labels, user clearances, zones, and overall system security.

The security analyst should only be allowed to assume this role under certain conditions. First, the analyst must be approved to perform administrative tasks. Second, they must authenticate again before switching to the role. Third, the role should only be used when making system or security changes, not for normal daily work. This helps reduce security risks by limiting when elevated privileges are used.

5. Create a report with the descriptions of your results. Include information about your thoughts on using Trusted Extensions.

I think Solaris Trusted Extensions are useful for environments that require strong security and clear separation between classification levels. The labeled zones make it easier to organize and control sensitive data, and the rules like no read up and no write down help prevent data leaks. I also like how roles are used to limit administrative access. However, I think it would take careful planning to set up correctly because everything depends on how the labels, zones, and roles are designed. Overall, I think it is a good system for government or intelligence environments where security is very important.

References

Oracle. *Trusted Extensions and the Oracle Solaris OS*.
https://docs.oracle.com/cd/E37838_01/html/E61029/txintro-4.html

Oracle. *Basic Concepts of Trusted Extensions*.
https://docs.oracle.com/cd/E26502_01/html/E29017/txintro-3.html

Oracle. *Planning for Security in Trusted Extensions*.
https://docs.oracle.com/cd/E37838_01/html/E61029/ovw-36.html